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Sec: 15

Assignment No-02

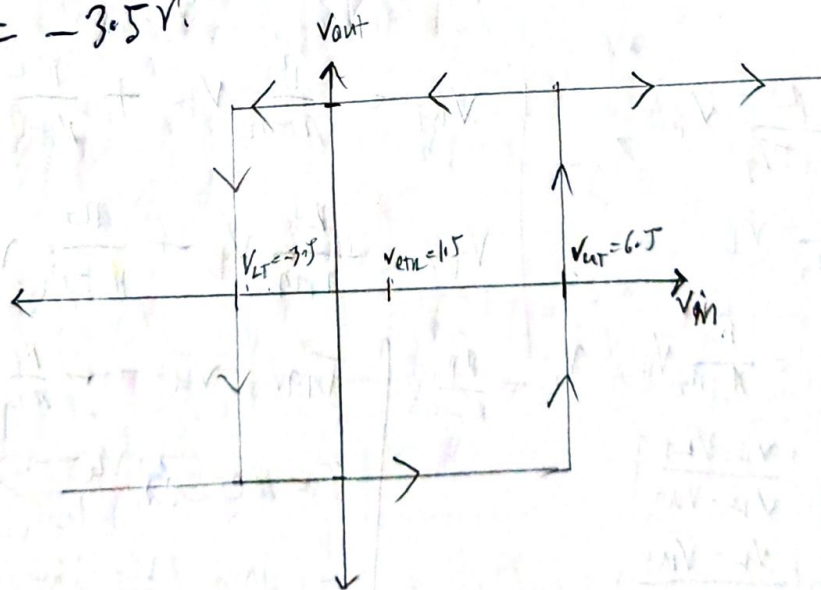
Ans to the Q No-1

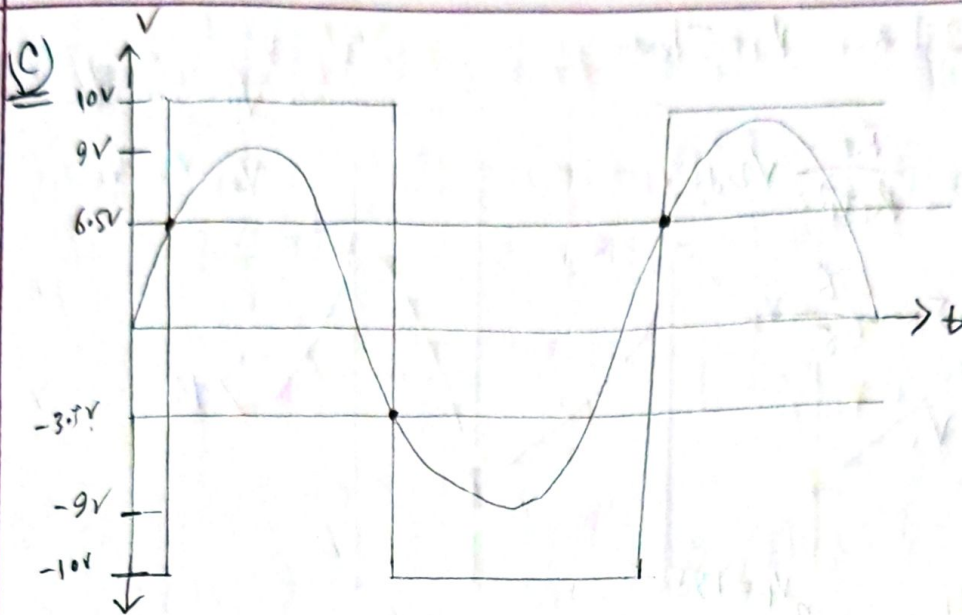
a) This non inverting schmitt trigger with applied voltage.

b) $V_H = 10V$, $V_L = -10V$, $V_{ref} = 1V$, $R_1 = 2k\Omega$, $R_2 = 2k\Omega$,

$$V_{out} = -\frac{R_1}{R_2} V_L + \frac{R_1 + R_2}{R_2} V_{ref}$$
$$= 6.5V.$$

$$V_{LT} = -\frac{R_1}{R_2} V_H + \frac{R_1 + R_2}{R_2} V_{ref}$$
$$= -3.5V.$$
$$V_{CTM} = \frac{6.5 - 3.5}{2}$$
$$= 1.5$$





$$V_H = 10V, V_L = -10V$$

$$V_{UT} = 6.5V,$$

$$V_{LT} = -3.5V$$

Ans to the Q No-2

(a) This is an inverting schmitt trigger with applied voltage.

Here, $V_H = 15V$, $V_L = -15V$, $V_{UT} = 10V$, $V_{LT} = 5V$, $V_S = \frac{V_{UT} + V_{LT}}{2}$

$$= \frac{15}{2} = 7.5V$$

(b) $V_{HYS} = V_{UT} - V_{LT} = 10 - 5 = 5V$

Now,

$$V_{HYS} = V_{UT} - V_{LT}$$

$$V_{HYS} = \frac{R_1}{R_1 + R_2} V_H - \frac{R_2}{R_1 + R_2} V_{UT} - \left(\frac{R_1}{R_1 + R_2} V_L - \frac{R_2}{R_1 + R_2} V_{LT} \right)$$

$$\frac{R_1}{R_1 + R_2} (V_H - V_L) = 5$$

$$\frac{R_1}{R_1 + R_2} = \frac{5}{30}$$

$$6R_1 = R_1 + R_2$$

$$5R_1 = R_2$$

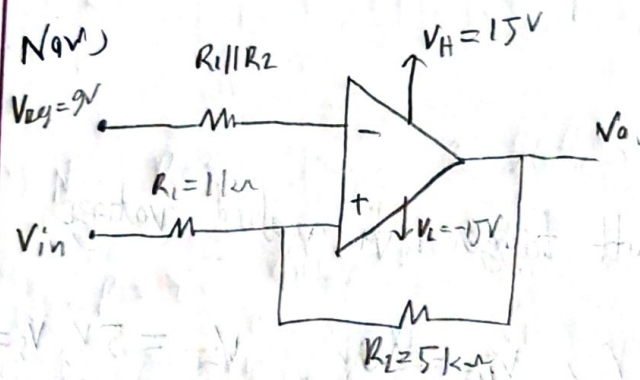
So, if $R_1 = 1k\Omega$, $R_2 = 5k\Omega$.

Now

$$V_{out} = V_3 = \frac{R_2}{R_1 + R_2} V_{ref}$$

$$\therefore V_{ref} = 7.5 \times \frac{6}{5}$$

$$\therefore V_{ref} = 9V$$



Ans to the Q No-3

(a) Here, $R_1 = R_2$, $V_H = 8V$, $V_L = -8V$,

$$f = 200Hz$$

$$\text{So } T = \frac{1}{200} \text{ Hz} = 5ms$$

$$\text{Now } D = \frac{T_1}{T} \times 100\%$$

$$\therefore T_1 = 0.5 \times 5 \times 10^{-3}$$

$$\therefore T_1 = 2.5ms, \text{ so } T_2 = 2.5ms$$

Now,

$$V_{UT} = \frac{R_1}{R_1 + R_2} V_H = \frac{1}{2} \times 8 = 4V$$

$$V_{LT} = \frac{R_1}{R_1 + R_2} V_L = \frac{1}{2} \times (-8) = -4V$$

Now,

$$T_1 = \tau \ln \left| \frac{V_H - V_{LT}}{V_H - V_{UT}} \right|$$

$$\therefore \tau = \frac{2.5 \times 10^{-3}}{\ln \left| \frac{8 + 4}{8 - 4} \right|}$$

$$\therefore \tau = \frac{2.5 \times 10^{-3}}{\ln \left| \frac{12}{4} \right|}$$

$$\therefore \tau = 2.276 \text{ ms.}$$

Now

$$\tau = R_x C_x$$

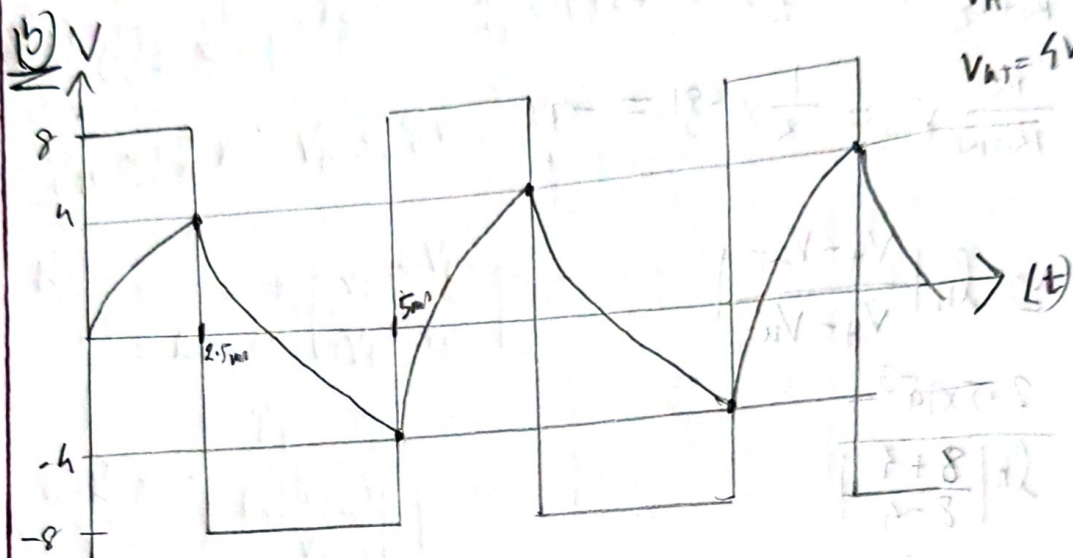
$$R_x = \frac{2.276 \times 10^{-3}}{100 \times 10^{-9}}$$

$$\therefore R_x = 22760 \Omega$$

So,

$$R_x = 22760 \Omega \text{ And } \tau = 2.276 \text{ ms.}$$

Ans



Here, $V_H = 8V$, $V_L = -8V$
 $V_{UT} = 4V$, $V_{LT} = -4V$

c) Now, $R_1 = 10k\Omega$, $R_2 = 80k\Omega$.

So, $p = 8$, and $V_L = -V_H$.

Now $V_{UT} = \frac{-V_L}{p} = \frac{-V_L}{8} = \frac{V_H}{8}$.

$V_{LT} = \frac{-V_H}{p} = -\frac{V_H}{8}$.

So, $T_1 = RC \left(\frac{V_{UT} - V_{LT}}{V_H} \right)$

$= RC \left(\frac{\frac{V_H}{8} + \frac{V_H}{8}}{V_H} \right)$

$= RC \left(\frac{2V_H}{8V_H} \right)$

$= \frac{RC}{4}$

$T_2 = RC \left(\frac{V_{LT} - V_{UT}}{V_L} \right)$

$= RC \left(\frac{-\frac{V_H}{8} - \frac{V_H}{8}}{-\frac{V_H}{8}} \right)$

$= RC \left(\frac{-2V_H}{-8V_H} \right)$

$= \frac{RC}{4}$

Now

$$T = T_1 + T_2 = \frac{RC}{4} + \frac{RC}{4} = \frac{2RC}{4} = \frac{RC}{2}$$

So

$$f = \frac{1}{T} = \frac{1}{\frac{RC}{2}} = \frac{2}{RC}$$

Ans

Ans to the Q No-4

(a) Here, $V_H = 8V$, $V_L = -8V$, $R_1 = 10k\Omega$, $R_2 = 40k\Omega$, $C = 5\mu F$, $R_i = 2k\Omega$

So

$$V_{UT} = -\frac{R_1}{R_2} V_L = 2V$$

$$V_{LT} = -\frac{R_1}{R_2} V_H = -2V$$

$$T_1 = RC \left(\frac{V_{UT} - V_{LT}}{V_H} \right) = 5ms$$

$$T_2 = RC \left(\frac{V_{LT} - V_{UT}}{V_L} \right) = 5ms$$

$$T = T_1 + T_2 = (5 + 5)ms = 10ms$$

$$f = \frac{1}{T} = \frac{1}{10 \times 10^{-3}} Hz$$

$$= 100Hz$$

So time period 10ms and frequency 100Hz.

Ans

b) $C_x = 50 \times 10^{-9} \text{ F}$, $T_1 = 0.25 \times 10^{-3} \text{ s}$, $V_{LT} = -10 \text{ V}$,

$V_{ut} = 5 \text{ V}$, $V_H = 8 \text{ V}$, $V_L = -18 \text{ V}$.

Now,

$$T_1 = RC \ln \left(\frac{V_H - V_{LT}}{V_H - V_{ut}} \right)$$

$$R_x = \frac{T_1}{C \ln \left(\frac{V_H - V_{LT}}{V_H - V_{ut}} \right)} = \frac{0.25 \times 10^{-3}}{50 \times 10^{-9} \times \ln \left(\frac{8 + 10}{8 - 5} \right)} = 2790.5 \Omega = 2.79 \text{ k}\Omega$$

~~Now, $R_1 = 1 \text{ k}\Omega$, $R_2 = 0.6 \text{ k}\Omega$~~

Now,

$$V_{ut} = \frac{R_1}{R_1 + R_2} V_H$$

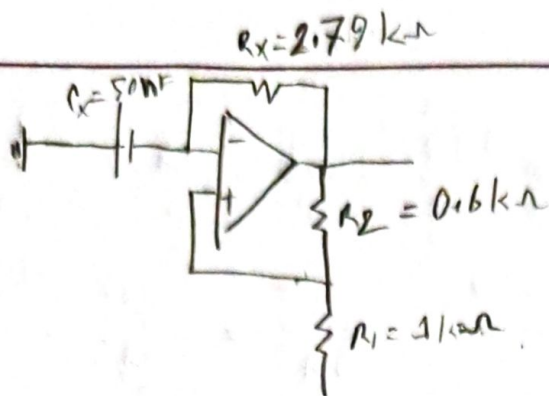
$$\frac{R_1}{R_1 + R_2} = \frac{5}{8}$$

$$5R_1 + 5R_2 = 8R_1$$

$$3R_1 = 5R_2$$

$$\frac{3}{5} R_1 = R_2$$

So, if $R_1 = 1 \text{ k}\Omega$, $R_2 = 0.6 \text{ k}\Omega$.



c) $R_x = 2.79 \text{ k}\Omega$, $C_x = 50 \text{ nF}$, $V_{UT} = 5 \text{ V}$, $V_{LT} = -10 \text{ V}$
 $V_H = 8 \text{ V}$, $V_L = -16 \text{ V}$, $R_1 = 1 \text{ k}\Omega$, $R_2 = 0.6 \text{ k}\Omega$.

so,

$$T_1 = R_x C_x \ln \left| \frac{V_H - V_{LT}}{V_H - V_{UT}} \right|$$

$$= 2.79 \times 10^3 \times 50 \times 10^{-9} \ln \left| \frac{8 + 10}{8 - 5} \right|$$

$$= 0.249 \text{ ms}$$

$$T_2 = R_x C_x \ln \left| \frac{V_L - V_{UT}}{V_L - V_{LT}} \right| = 2.79 \times 10^3 \times 50 \times 10^{-9} \times \ln \left| \frac{-16 - 5}{-16 + 10} \right|$$

$$= 0.174 \text{ ms}$$

so,

$$T = T_1 + T_2 = (0.249 + 0.174) \text{ ms} = 0.423 \text{ ms}$$

so, $f = \frac{1}{T} = \frac{1}{0.423 \times 10^{-3}} = 2364.1 \text{ Hz}$

$$D = \frac{T_1}{T_1 + T_2} \times 100\% = \frac{0.249}{0.423} \times 100\% = 58.86\%$$